

Designing and Analysis of Algorithms

Course Code: ECS 5101/CS514

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- **Lecture 12**

TREES

- > It is a non-linear data structure.
- > It is mainly used to represent data containing a hierarchical relationship between element e.g. table of contents.
- > Imp: No loop should be formed.

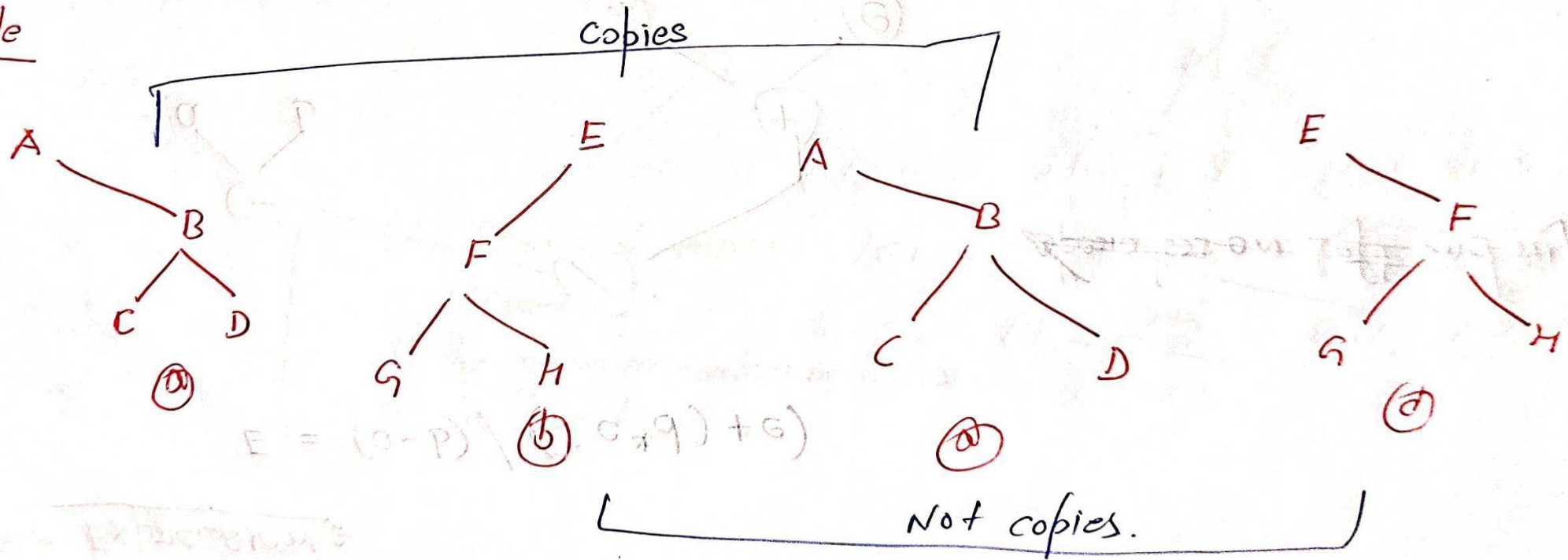
* Binary Trees :

- A binary tree T is defined as a finite set of elements, called nodes, such that
- T is empty (called the null tree or empty tree) or
 - T contains a distinguished node R , called the root of T , and the remaining nodes of T forms an ordered pair of disjoint binary trees T_1 and T_2 .

Note:

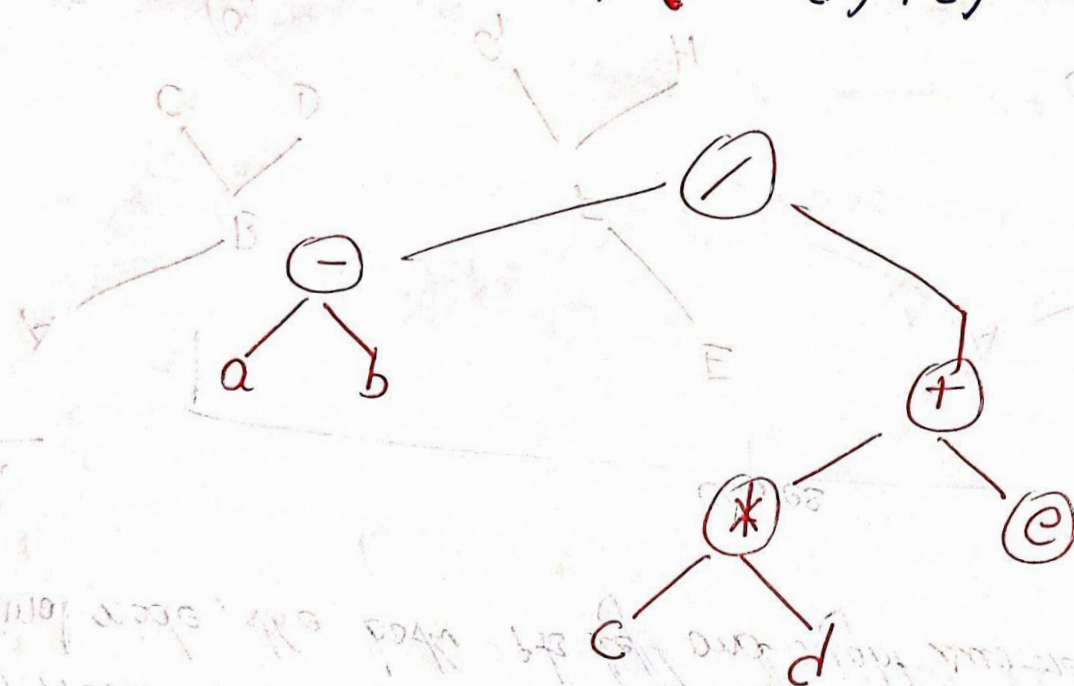
- 1) Any node N in a binary tree T has either 0, 1 or 2 successors.
- 2) The node with no successors are called terminal nodes.
- 3) Every node N of T contains a left and a right subtree. Moreover, if N is a terminal node, the both its left and right subtrees are empty.

Example



Algebraic Expression :-

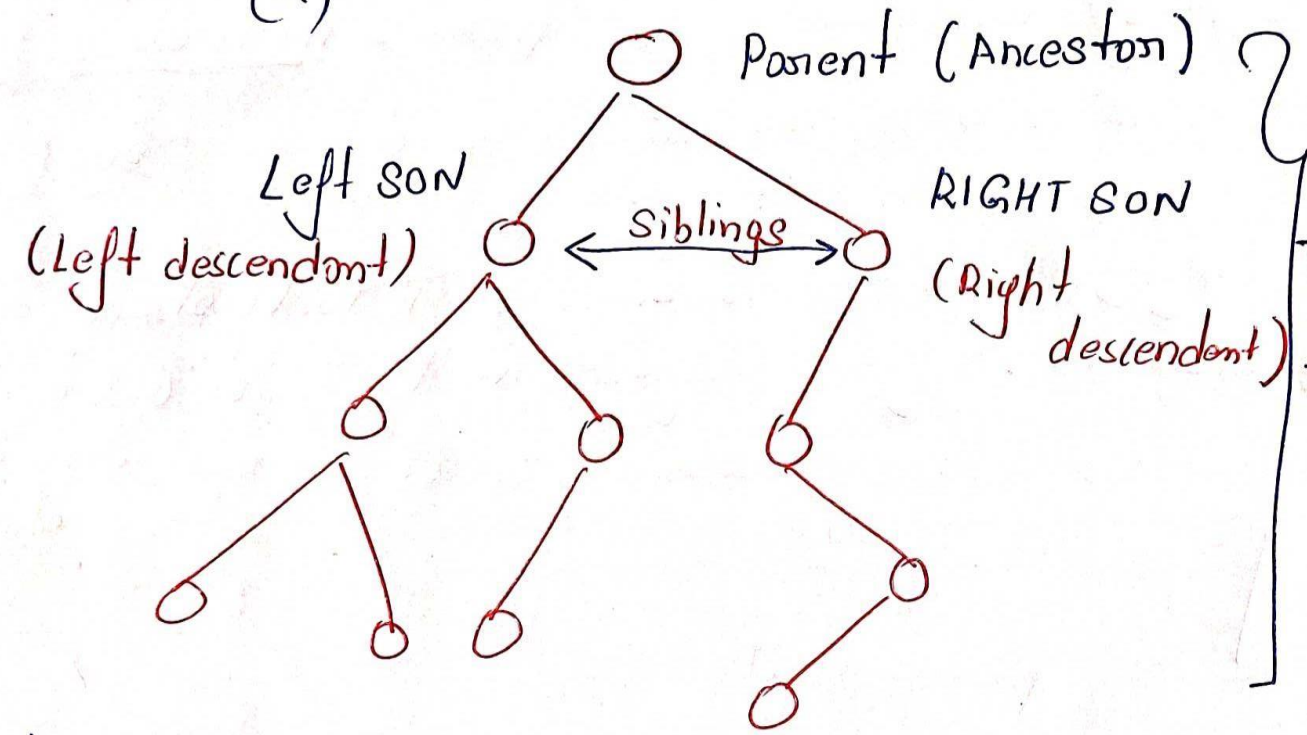
$$E = (a-b) / ((c*d) + e)$$



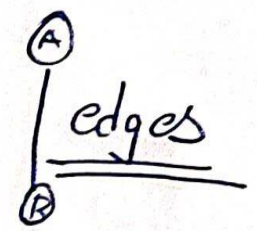
~~operators on left and right~~

Terminology :-

(a)

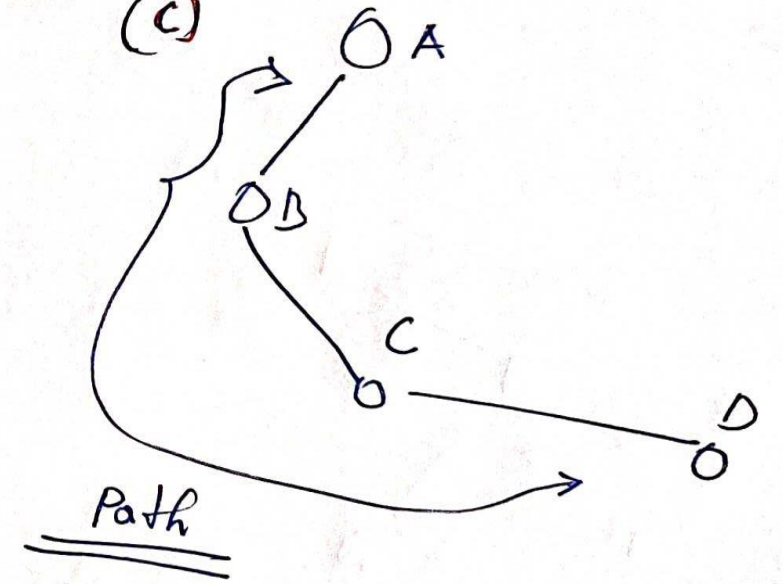


(b)

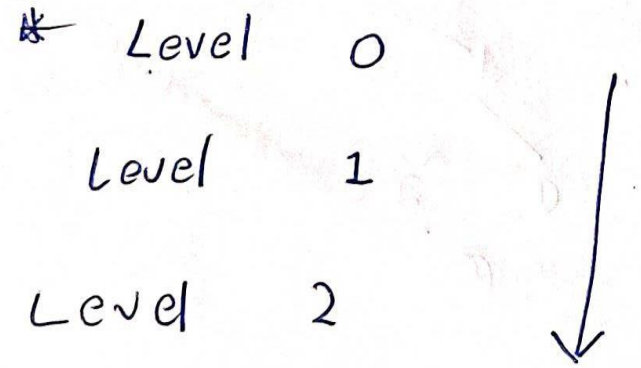


TREE

(c)



(d)



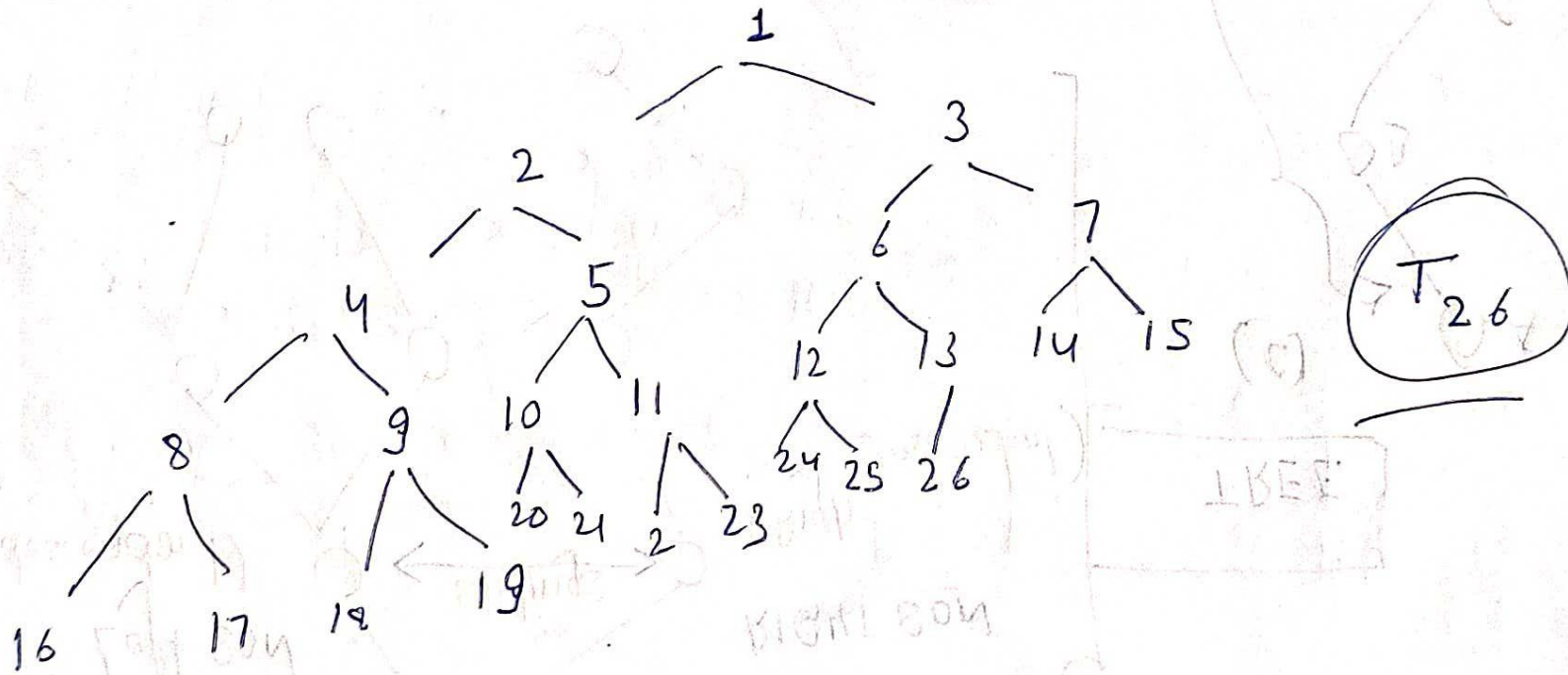
(e) Height of tree:
(Depth)

is the maximum number of nodes
in any branch of T.

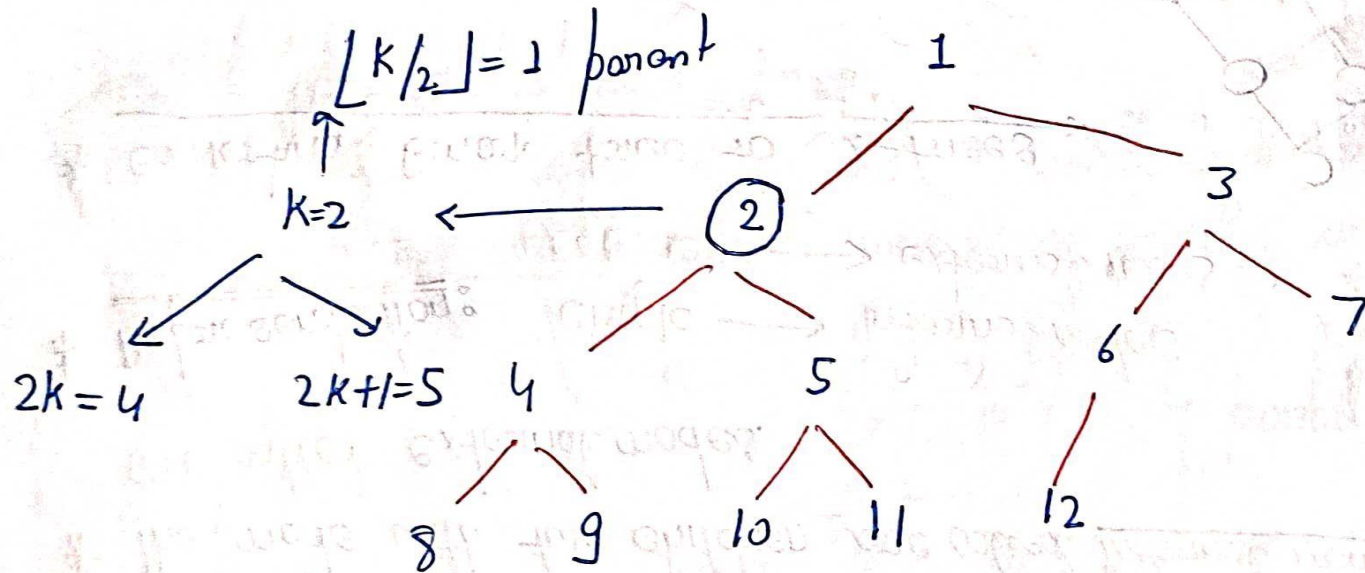
Complete Binary Trees:

* The tree T is said to be complete ^(P1) if all its levels, except possibly the last, have the maximum number of possible nodes, i.e. 2.

* ^(P2) if all the nodes at the last level appear as far left as possible.



The left and right children of node (K) are, respectively, $2 \cdot K$ and $2 \cdot K + 1$ and the parent of K is the node $\lfloor K/2 \rfloor$.



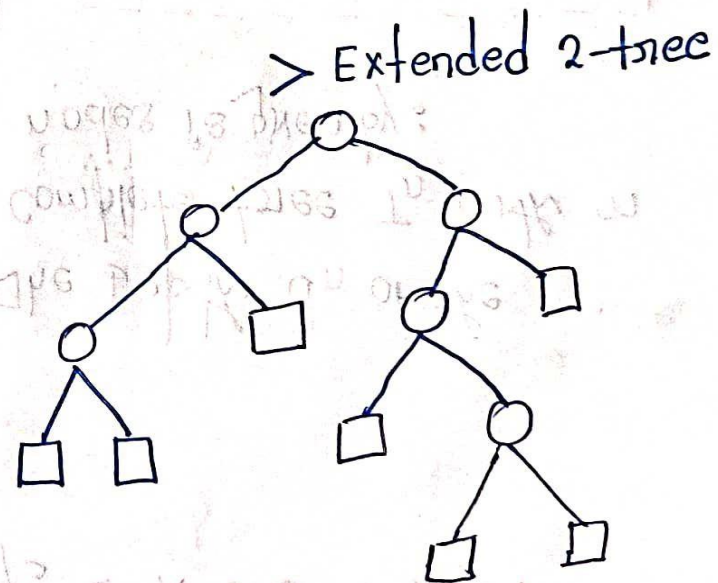
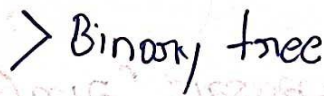
The Depth d_n of the Complete tree T_n with n nodes is given by:

$$d_n = \lfloor \log_2 n + 1 \rfloor$$

Example: IF the complete tree T_n has $n = 1\,000\,000$ nodes, then its depth

$$D_n = 21$$

⑧



Representing Binary Trees in Memory :

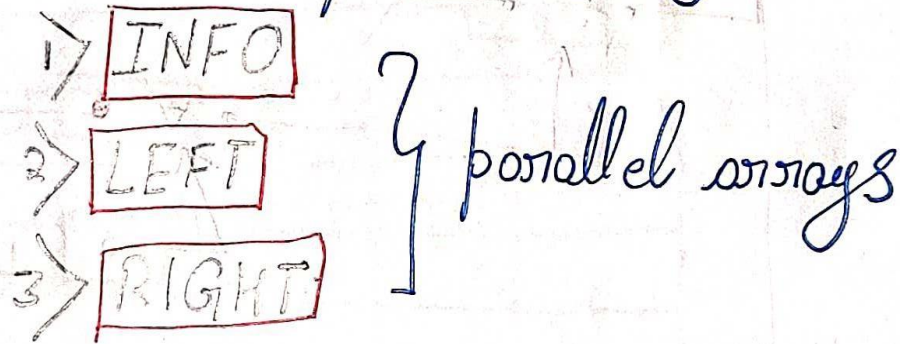
⑨

① Linked represent.

② Sequential representation using single array.

> Let T denotes a binary tree.

> Unless otherwise stated or implied, T will be maintained in memory by means of a linked representation which uses three parallel arrays

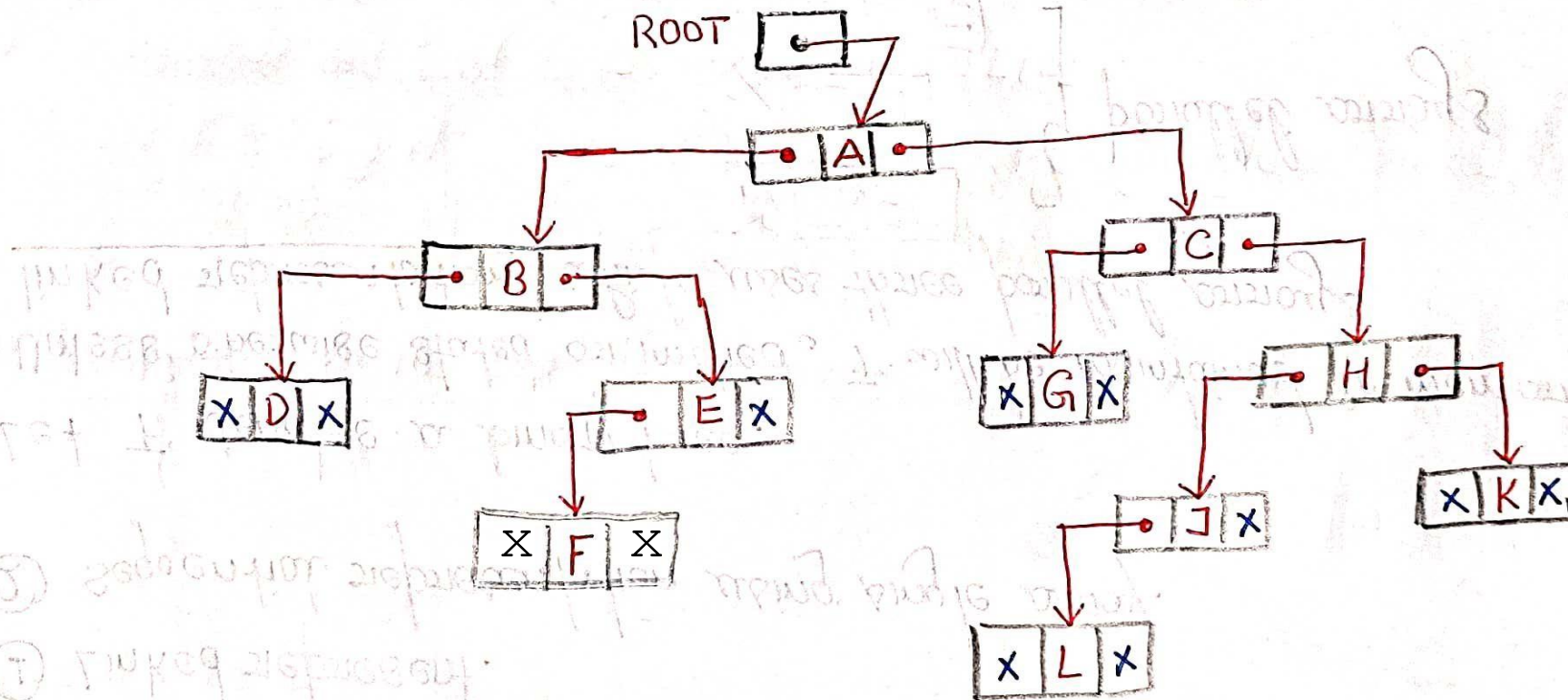


and one pointer variable "ROOT"

Each node (N) of (T) will correspond to a location (K) such that:

(10)

- (1) $INFO[K]$ contains the data of the node N
- (2) $LEFT[K]$ contains the location of the left child of node N
- (3) $RIGHT[K]$ contains the location of the right child of node N .



ROOT
5

AVAIL
8

	INFO	LEFT	RIGHT
1	K	0	0
2	C	3	6
3	G	0	0
4		14	
5	A	10	2
6	H	17	1
7	L	0	0
8		8 9	
9		4	
10	B	18	13 + 1
11		19	
12	F	0	0
13	E	12	0
14		15	
15		16	
16		17 11	
17	J	7	0
18	D	0	0
19		20	
20		0	

Sequential Representation of Binary Trees:

- > Suppose T is a binary tree that is complete or nearly complete.
- > An effective way to maintain T in memory called the sequential representation of T .
- > This representation uses only a single linear array $TREE$ as follows:

(a) The root R of T is stored in $TREE[1]$

(b) If a node N occupies $TREE[K]$, then its left child is stored in $TREE[2 * K]$ and its right child is stored in $TREE[2 * K + 1]$

* NULL is used to indicate an empty subtree. Particular, $TREE[1] = \text{NULL}$ indicate that the tree is empty.